Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 85100

Part A: Each question is worth 3 points

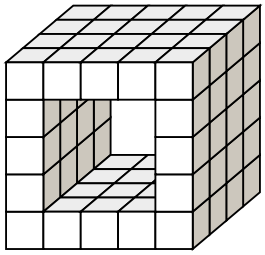
1. The digits from 1 to 9 are written on 9 cards. Mia has 4, 8 and 9; Nina has 6, 5 and 2; Lily has 1, 3 and 7. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.

Who **cannot** get 20 as the result?

A. Mia ***B.*** Nina ***C.*** Lily

D. All can get 20 ***E.*** Nina and Lily

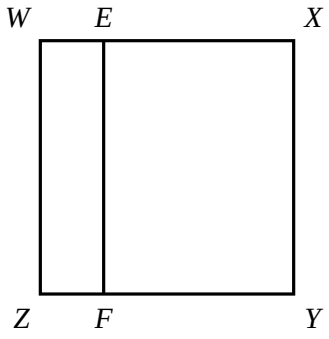
2. Sara built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Sara use?

A. 64 ***B.*** 65 ***C.*** 67 ***D.*** 80 ***E.*** 100

3. *WXYZ* is a square. Its side is 12 cm. *WEFZ* is a rectangle. Its shorter side is 3 cm.



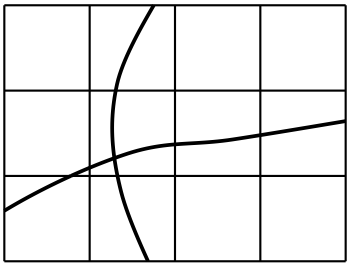
How many centimetres is the perimeter of the square *WXYZ* larger than that of the rectangle *WEFZ*?

A. 36 cm ***B.*** 33 cm ***C.*** 30 cm ***D.*** 24 cm ***E.*** 18 cm

4. Noah, Mia, Chloe and Tom each have a different pet: a goldfish, a rabbit, a dog and a hamster. Tom has a pet that barks. Mia has a pet that lives in water. Chloe doesn't like hamsters. Who has a dog?

A. Tom ***B.*** Noah ***C.*** Mia
D. Chloe ***E.*** Impossible to determine

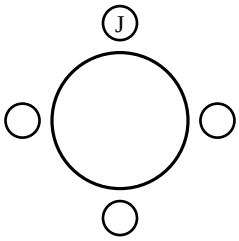
5. Mia drew some lines on squared paper. Here is the whole picture.



Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

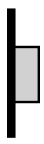
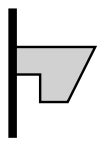
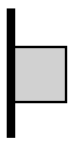
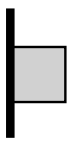
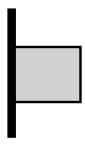
6. Four friends, John, Ben, Anna and Felix, go to a restaurant and sit at a table with 4 chairs. John always sits on the same chair (marked J).



In how many different ways can the friends sit around the table?

A. 3 ***B.*** 4 ***C.*** 6 ***D.*** 9 ***E.*** 24

7. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments.
Which picture **cannot** be right unless the flag is torn?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

8. Mia made a tunnel out of identical cubes (fig. 1). Then Mia took it apart and built a complete pyramid (fig. 2) with some of the cubes. Nothing is hollow inside.

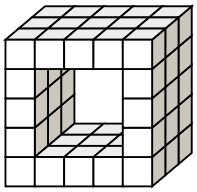


fig. 1

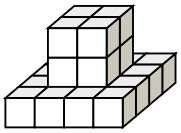


fig. 2

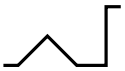
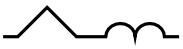
How many cubes from the tunnel were **not** used for the pyramid?

A. 24 ***B.*** 25 ***C.*** 28 ***D.*** 36 ***E.*** 40

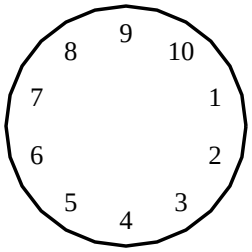
Part B: Each question is worth 4 points

9. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A.  ***B.*** ***C.***  ***D.*** ***E.***

10. A round plate with the numbers 1 to 10 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



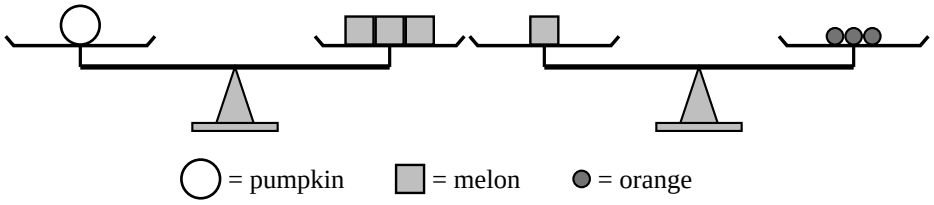
What is the largest of these sums?

A. 18 ***B.*** 27 ***C.*** 28 ***D.*** 29 ***E.*** 32

11. The odometer of a car shows 6335 km. Some of its digits are the same.
How many kilometres ago did the odometer last show a number with all different digits?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 16

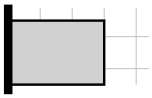
12. The two balances are level. One pumpkin weighs as much as 3 melons, and one melon weighs as much as 3 oranges.



How many oranges weigh as much as 1 pumpkin?

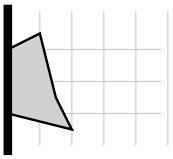
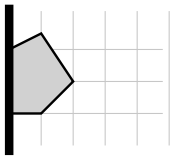
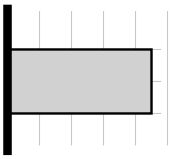
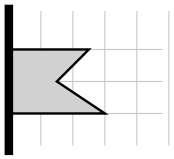
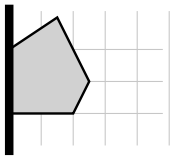
A. 3 ***B.*** 6 ***C.*** 8 ***D.*** 9 ***E.*** 10

13. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Karla draws the shadow at five different moments. Spread out flat, the flag is 3 squares wide and 2 squares tall (see the picture).

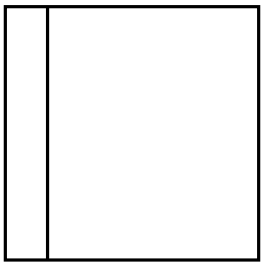


the flag, flat

Which shadow **cannot** be right?

A. 
B. 
C. 
D. 
E. 

14. A square has perimeter 20 cm. It is cut along straight lines into 2 rectangles as shown.



What is the sum of the perimeters of all 2 rectangles?

A. 30 cm ***B.*** 33 cm ***C.*** 35 cm ***D.*** 36 cm ***E.*** 40 cm

15. Alex has the cards 2, 3 and 9. Alex must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Alex **not** get?

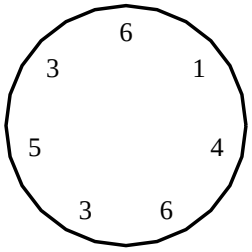
A. 33 ***B.*** 30 ***C.*** 29 ***D.*** 25 ***E.*** 24

Part C: Each question is worth 5 points

16. Julien, Leo, Lucas and Omar each have a different pet: a goldfish, a hamster, a cat and a canary. Omar doesn't like hamsters. Leo has a bird. Julien has a pet that lives in water.
Which statement is **not** true?

A. Julien has a goldfish ***B.*** Lucas has a hamster ***C.*** Omar has a cat
D. Leo has a canary ***E.*** Leo has a hamster

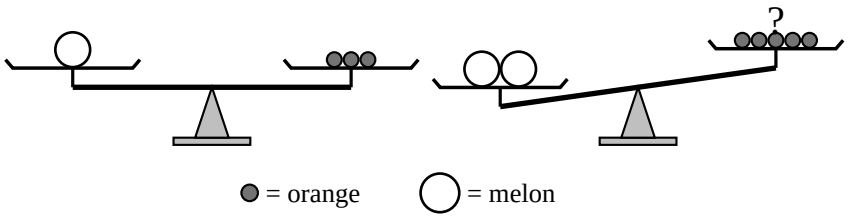
17. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

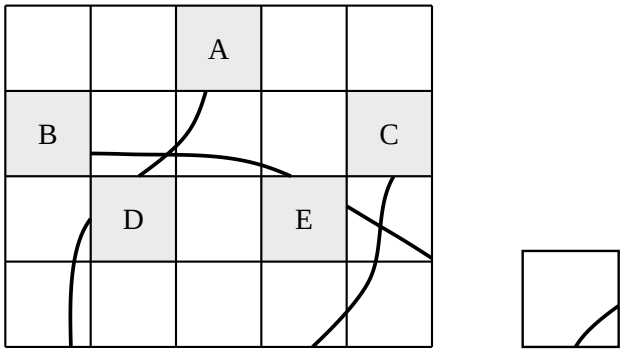
18. The first balance is level. On the second balance there are 2 melons on the left and 5 oranges on the right.



How many oranges must be added to the right plate to make it level?

A. 1 ***B.*** 2 ***C.*** 4 ***D.*** 6 ***E.*** 11

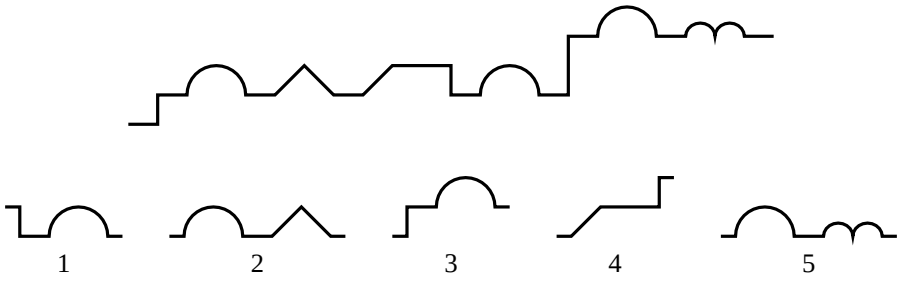
19. A cat played with balls of wool on squared paper. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

20. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 4 ***E.*** 5

Answer Key

1.	**A**	**2.**	**A**	**3.**	**E**	**4.**	**A**	**5.**	**D**	**6.**	**C**
7.	**B**	**8.**	**A**	**9.**	**A**	**10.**	**C**	**11.**	**B**	**12.**	**D**
13.	**C**	**14.**	**A**	**15.**	**B**	**16.**	**E**	**17.**	**B**	**18.**	**A**
19.	**C**	**20.**	**D**								

Solutions

1. A (Mia) — Nina: (6 – 2) × 5 = 20; Lily: 3 × 7 – 1 = 20. Mia: with 4, 8 and 9 nothing gives 20 (the closest are 18 = 8 × 9 ÷ 4 and 21 = 4 + 8 + 9). [set-03/by_level/Grade_03-04/2002 Q21 · who_cannot]

2. A (64) — The front has 5 × 5 – 3 × 3 = 16 cubes. The tunnel is 4 cubes long, so 16 × 4 = 64. [set-03/by_level/Grade_03-04/2002 Q20 · count_tunnel]

3. E (18 cm) — Square: 4 × 12 = 48 cm. Rectangle: 3 + 12 + 3 + 12 = 30 cm. Difference 48 – 30 = 18 cm. [set-03/by_level/Grade_03-04/2002 Q6 · square_minus_strip]

4. A (Tom) — The only possible matching: Noah – hamster; Mia – goldfish; Chloe – rabbit; Tom – dog. [set-03/by_level/Grade_03-04/2002 Q18 · who_has]

5. D — Find each square in the picture by the places where its lines reach the edges. Squares A, B, C, E can be found; square D is a piece mirrored and turned, so it is not in the picture like that. [set-03/by_level/Grade_03-04/2002 Q1 · not_a_piece]

6. C (6) — John's chair is fixed. The other 3 friends fill the other chairs: 3 × 2 × 1 = 6 ways. [set-03/by_level/Grade_03-04/2002 Q22 · fixed_seat]

7. B — The side sewn to the pole always shows at its full length. In picture B part of the pole is bare, so a piece of the flag is missing. [set-03/by_level/Grade_03-04/2002 Q15 · torn_flag]

8. A (24) — Tunnel: 16 cubes in the front, 3 long: 16 × 3 = 48. Pyramid: 4 × 4 + 2 × 2 × 2 = 24. Left over: 48 – 24 = 24. [set-03/by_level/Grade_03-04/2002 Q20 · tunnel_to_pyramid]

9. A — Only this piece is found in the line. The others don't fit — B: it is a mirror image of a part of the line; C: it is a mirror image of a part of the line; D: its slope goes the wrong way; E: it has a round dome where the line has a pointed roof. [set-03/by_level/Grade_03-04/2002 Q7 · can_belong]

10. C (28) — All the numbers add up to 55. Two consecutive numbers that add up to 55: 27 + 28 = 55. The largest is 28. [set-03/by_level/Grade_03-04/2002 Q11 · largest_sum]

11. B (6) — Counting down from 6335, the first number with all different digits is 6329: 6335 – 6329 = 6. [set-03/by_level/Grade_03-04/2002 Q24 · last_time]

12. D (9) — One pumpkin = 3 melons = 3 × 3 = 9 oranges. [set-03/by_level/Grade_03-04/2002 Q9 · chain]

13. C — Turning or folding can only make the flag look narrower. Picture C reaches more than 3 squares away from the pole, which is wider than the flag. [set-03/by_level/Grade_03-04/2002 Q15 · too_wide]

14. A (30 cm) — Side = 20 ÷ 4 = 5 cm. There is 1 cut; each cut is 5 cm long and is counted twice (once for each piece next to it): 20 + 1 × 2 × 5 = 30 cm. The widths of the pieces don't matter! [set-03/by_level/Grade_03-04/2002 Q6 · pieces_total]

15. B (30) — 2 × (3 + 9) = 24; 3 × 9 – 2 = 25; 2 + 3 × 9 = 29; (2 + 9) × 3 = 33. But with 2, 3 and 9 nothing gives 30 (the closest are 29 = 2 + 3 × 9 and 33 = (2 + 9) × 3). [set-03/by_level/Grade_03-04/2002 Q21 · which_cannot]

16. E (Leo has a hamster) — The only possible matching: Julien – goldfish; Leo – canary; Lucas – hamster; Omar – cat. [set-03/by_level/Grade_03-04/2002 Q18 · which_not_true]

17. B (2) — Each half must have 14. The ways: 1+4+6+3 | 5+3+6; 6+3+5 | 3+6+1+4. [set-03/by_level/Grade_03-04/2002 Q11 · count_ways]

18. A (1) — One melon = 3 oranges, so the left plate is worth 2 × 3 = 6 oranges. The right plate has 5, so add 6 – 5 = 1. [set-03/by_level/Grade_03-04/2002 Q9 · what_to_add]

19. C (gap C) — The square has top: no line, right: 1 line, bottom: 1 line, left: no line. Only gap C has lines reaching it at exactly those places. [set-03/by_level/Grade_03-04/2002 Q1 · which_gap]

20. D (4) — Pieces 1, 2, 3, 5 fit. Not: 4 (it steps up where the line steps down). [set-03/by_level/Grade_03-04/2002 Q7 · how_many_fit]